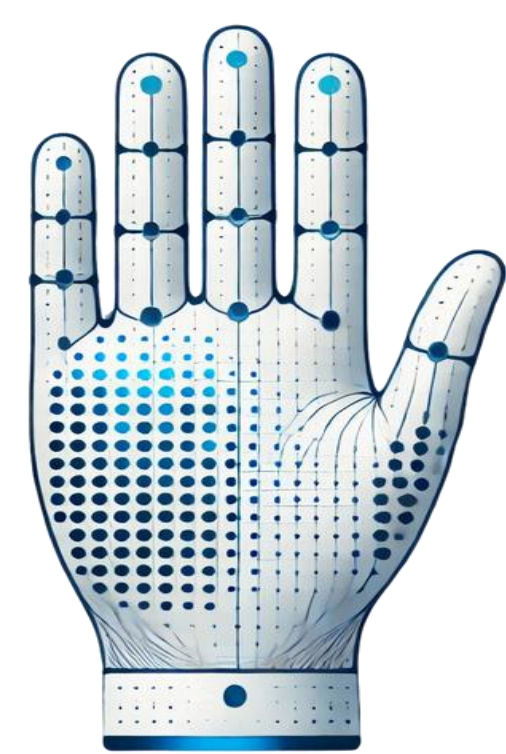




HDSense Glove: Redesigning Tactile Sensing

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Abstract

The **HDSense Glove** is a portable tactile sensing device designed for real-time haptic feedback in prosthetics, robotics, and industrial automation. It combines a **1024 high-density flexible sensor grid**, compressed sensing, and machine learning algorithms to optimize data processing and feedback accuracy.

By addressing challenges like high data load, cost, and portability, the glove delivers:

- **18x faster processing** than traditional raster scanning.
- **88% detection accuracy** with significant data reduction.
- **Seamless wireless transmission** for real-time performance.

The lightweight, flexible design ensures comfort and adaptability, making the **HDSense Glove** a scalable and efficient solution for precision tactile sensing.

Introduction

The ability to mimic the sense of touch is a game-changer in fields like **prosthetics**, **robotics**, and **human-machine interaction**. Existing tactile sensing systems often struggle with **high data loads**, **bulky designs**, and **limited adaptability**, making them unsuitable for dynamic real-world applications. The **HDSense Glove** addresses these challenges through an innovative design for reconstructing high-fidelity signals. A prototype of the glove is shown below



Figure 1: Prototype of **HDSense Glove**

As illustrated in the schematic below, the HDSense Glove follows a streamlined workflow.

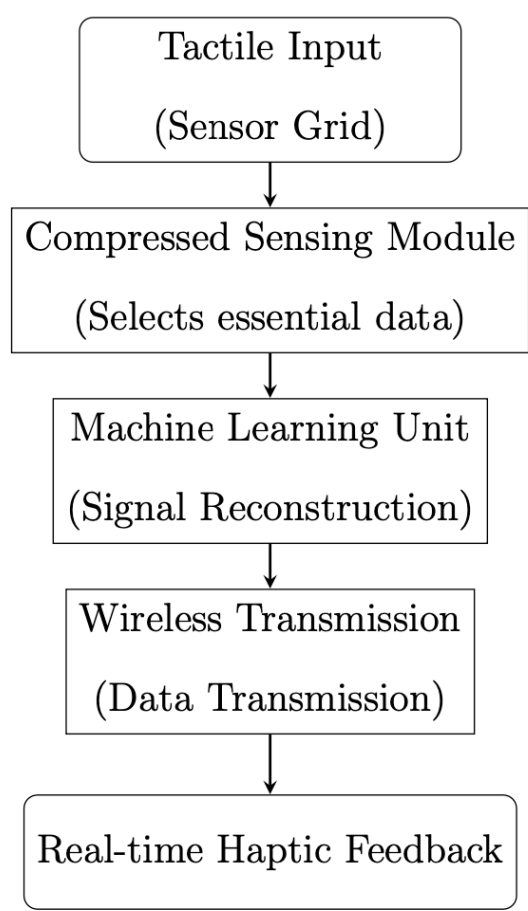
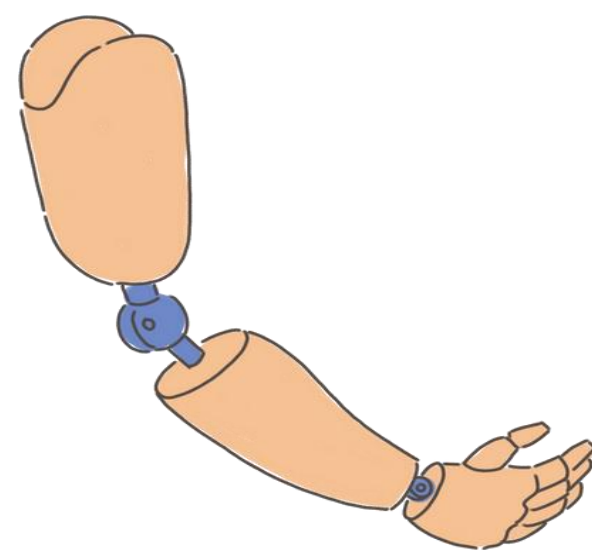


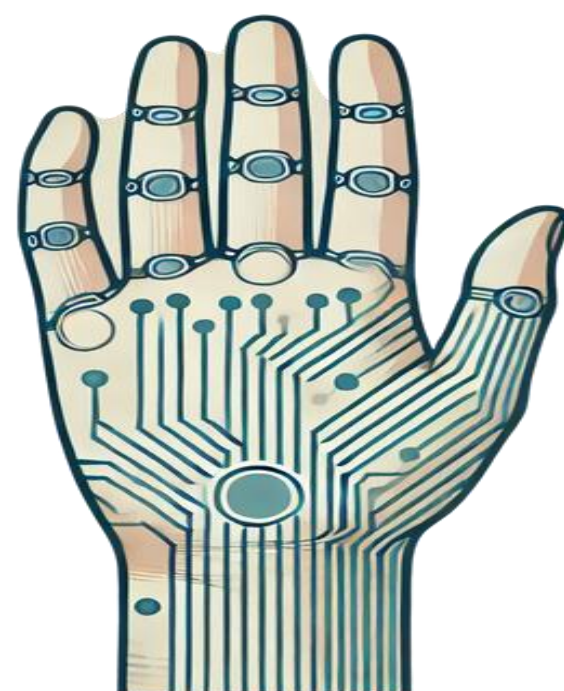
Figure 2: Workflow Diagram

The HDSense Glove provides a **portable**, **scalable**, and **cost-effective** solution, redefining what's possible in precision tactile sensing.

Application



Upper Limb Prosthetics
Enhancing precision and feedback for prosthetic control.



Electronic Skin
Enabling advanced touch-sensitive surfaces for robotics.



Smart Sports Equipment:
Revolutionizing performance tracking with real-time tactile data.

Results

Prototype testing validated the **HDSense Glove's** ability to deliver accurate and efficient tactile sensing. Key findings include:

- High accuracy:** The glove achieved over **88% detection accuracy**, ensuring precise feedback.
- 18x faster data processing:** Compared to raster scanning techniques, the glove provides real-time performance, critical for dynamic applications.
- Improved latency:** Compressed sensing algorithms reduced latency to about **20ms**, much faster than other products.

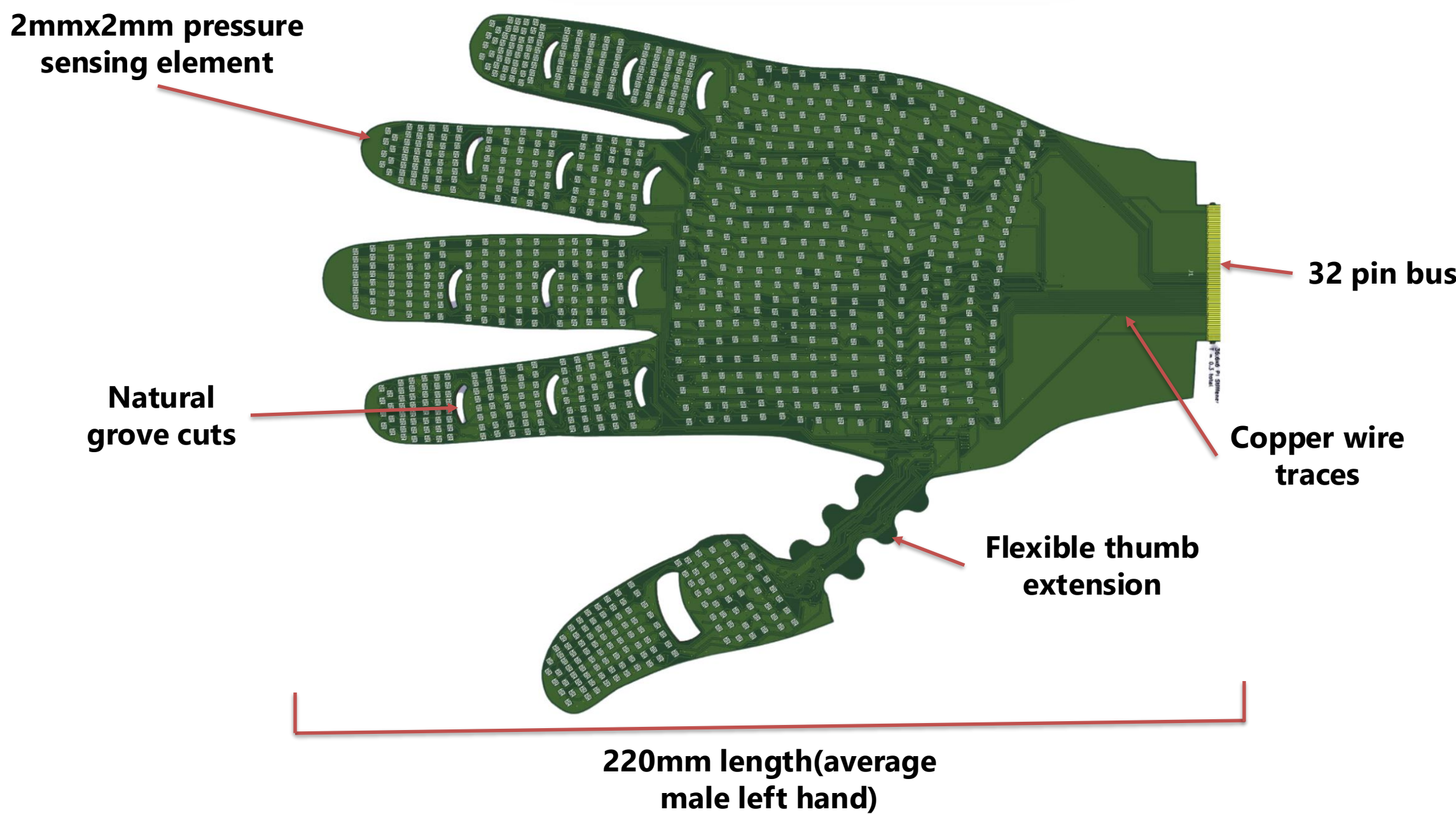


Figure 3: Diagram PCB Design on **KiCad** software

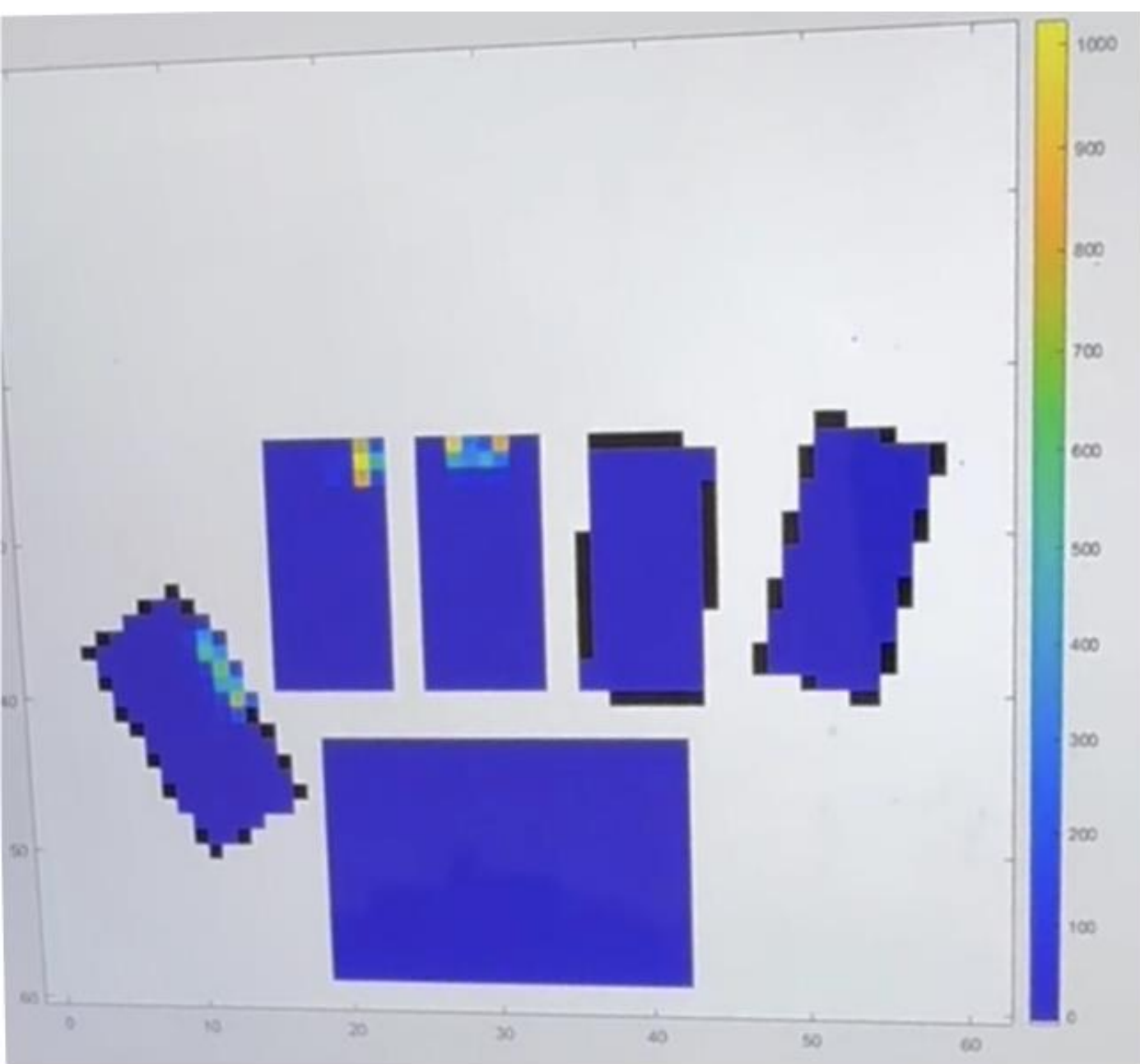


Figure 4: Sensor mapping on **MATLAB GUI**

Competitive Edge

The following table highlights the **HDSense Glove's competitive edge** in comparison to existing tactile sensing systems, demonstrating its superior performance across key metrics.

FEATURE/ METRIC	STAG GLOVE(MIT)	ELECTRONIC SKIN SYSTEMS	SMART SPORTS GLOVE	HDSENSE GLOVE
Sensor Density	548	Low Density	Low Density	1024
Frame Rate	7.3Hz	<100Hz	<100Hz	1000Hz
Latency	High	High	Medium	20ms
Energy Consumption	Medium	Low	Low	Low
Resolution	Medium	Low	Low	Very High

The results clearly showcase how the **HDSense Glove** outperforms its competitors with higher sensor density, faster frame rates, and minimal latency, offering a scalable and efficient solution for real-world applications in prosthetics, robotics, and industrial automation.

Conclusions

The **HDSense Glove** revolutionizes tactile sensing with its high-density sensor grid, compressed sensing, and machine learning, delivering real-time, accurate feedback while addressing challenges of speed and scalability.

With its proven performance and clear competitive edge, the glove sets the stage for smarter, more accessible solutions in prosthetics, robotics, and beyond, redefining the future of human-machine interaction.

Additional Questions?

To learn more about our work, scan the QR Code to visit the lab website, or click the LinkedIn and email icons to connect with us directly.

